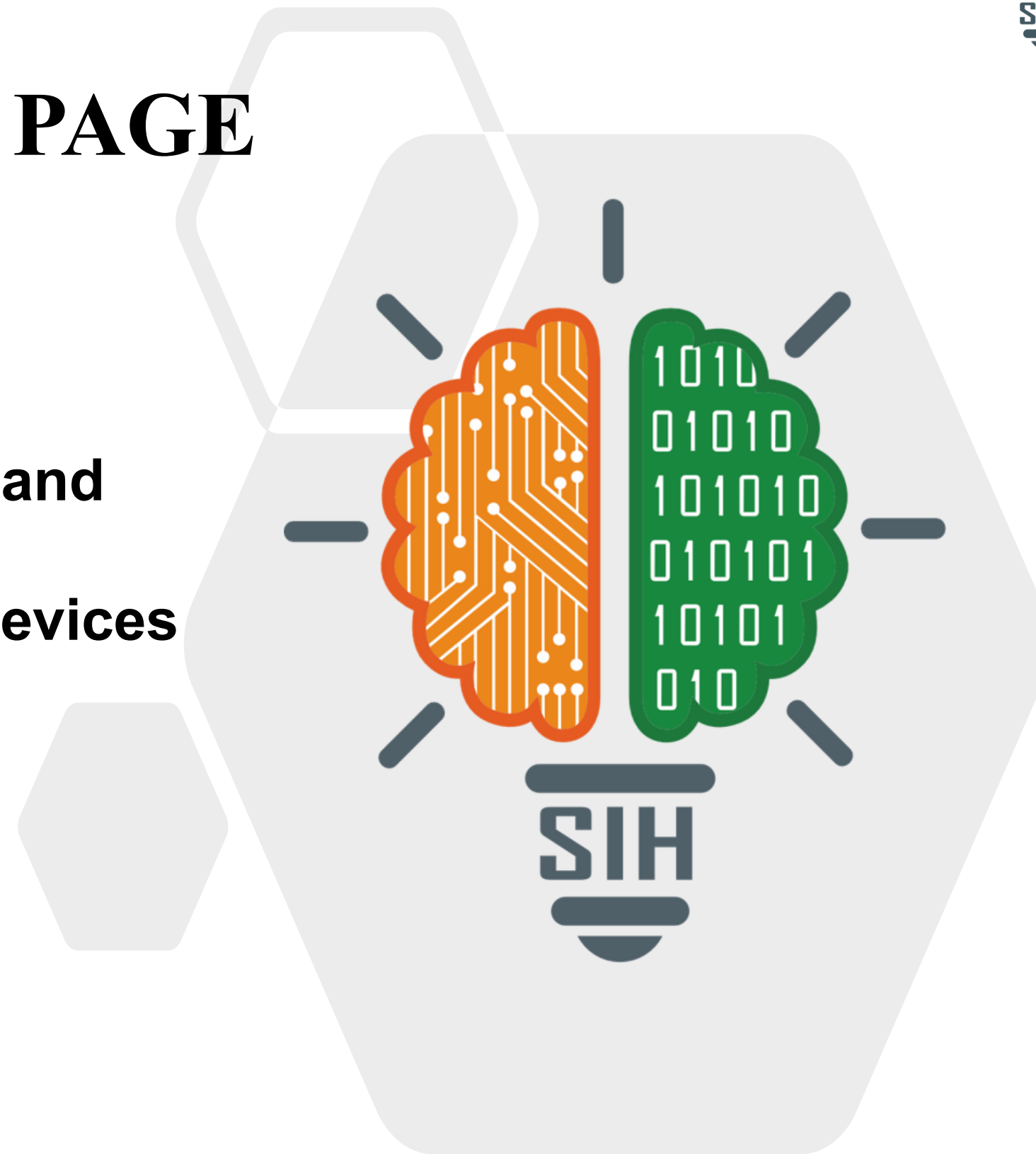


TITLE PAGE

- **Problem Statement ID – 26172**
- **Problem Statement Title- Low Latency and Efficient Voice Activator for Edge Devices**
- **Theme - Smart Automation**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name (Registered on portal) - TRINETRA**



• Proposed Solution (Describe your Idea/Solution/Prototype)

1. Detailed Explanation of the Proposed Solution

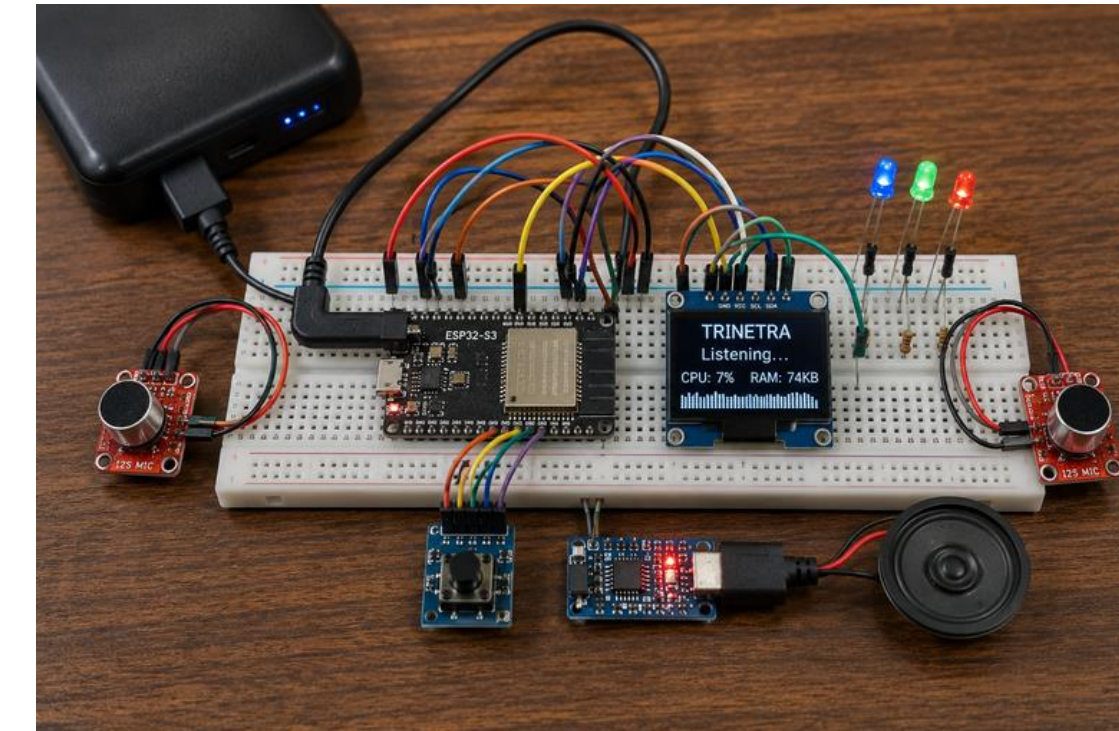
- **Edge Device:** ESP32-S3 based low-cost edge device with dual I2S microphones for continuous local listening.
- **Adaptive Cascaded Wake Engine (ACWE):** Three-level detection – Energy Gate → VAD → TinyML KWS. Heavy processing activates only when required.
- **Dynamic Sampling Strategy (DSS):** Adjusts processing quality and window size based on environment and activity for optimal efficiency.
- **Smart Offline Queue:** Stores queries, audio, and events locally in flash (max 5) with metadata and uploads sequentially when internet is available.
- **Privacy First:** All ambient audio is processed locally. Only user query (after wake-word) is sent to cloud.
- **Outputs:** Response via speaker, status on OLED, and full details on web dashboard.

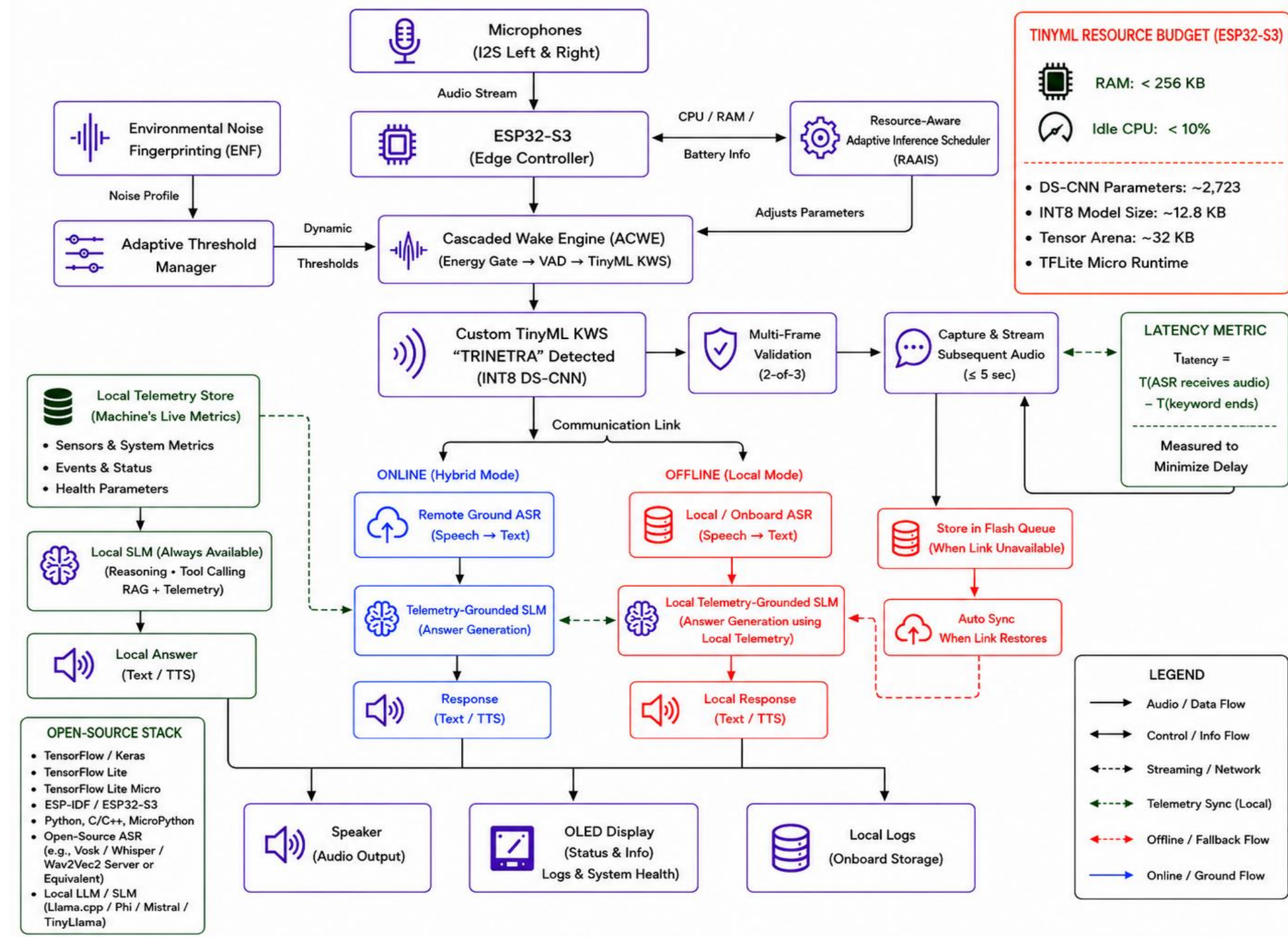
2. How It Addresses the Problem?

- **Low Latency:** Event-driven multi-level detection ensures instant wake response (< 1s). Post-wake audio is streamed immediately to ASR.
- **Low Resource Usage:** ACWE + RAAIS keeps idle CPU <10% and RAM <256 KB using adaptive scheduling and INT8 DS-CNN.
- **High Accuracy:** Noise-aware thresholds + multi-frame (2-of-3) validation significantly reduces false activations in real-world environments.
- **Offline Resilience (Hybrid Mode):** Works without internet using local telemetry, onboard ASR/SLM. Flash queue ensures no query or event is lost; auto-sync when link restores.
- **Privacy & Security:** Continuous ambient audio never leaves the device; only captured query audio is transmitted. Machine-specific telemetry is isolated.
- **Easy Customization:** Wake word, thresholds, noise profile, and system behavior can be updated as per user needs.

3. Innovation and Uniqueness of the Solution

- **RAAIS-ACWE Engine:** First-of-its-kind combination of Resource-Aware Adaptive Inference Scheduler (RAAIS) with Adaptive Cascaded Wake-Word Engine (ACWE).
- **Environmental Noise Fingerprinting (ENF):** Creates noise profile (quiet, fan, traffic, crowd, etc.) and adapts thresholds for robust activation.
- **Self-Adaptive Intelligence:** Dynamically adapts to CPU, RAM, noise and connectivity in real-time for maximum reliability and efficiency.
- **End-to-End Automation:** From listening → detection → capture → ASR/SLM → response with minimal user interaction.
- **Low Cost, High Efficiency:** Runs on affordable ESP32-S3 hardware with simple peripherals (mic, speaker, OLED, LEDs).
- **Scalable & Extensible:** Modular firmware and architecture supports future models, features and multi-machine deployment.





ARCHITECTURE

- Two I2S microphones for noise-aware voice capture.
- Cascaded wake-word pipeline (Energy Gate → VAD → TinyML KWS) on ESP32-S3.
- Adaptive thresholding using ENF and RAAIS for high accuracy (<10% idle CPU, <256 KB RAM).
- After wake word, capture (max 5 sec) and send to remote ASR server for STT → SLM → TTS.
- If link unavailable, store in flash (offline queue) and auto-sync when connection returns.
- Real-time status and responses on OLED, speaker and web dashboard.

COMPONENTS

- Edge Device: ESP32-S3 with I2S mics, OLED, speaker, status LEDs.
- TinyML Engine: Adaptive Cascaded Wake-Word Engine (Energy Gate → VAD → KWS).
- Cloud AI: Remote ASR (STT), SLM (Answer Generation) and TTS.
- Database: Stores user queries and system logs (remote server).
- Offline Module: Flash-based queue with retry and checksum for reliable sync.
- Dashboard: Web interface to monitor device status and analytics.

TECHNOLOGY STACK

HARDWARE	EDGE AI	CUSTOM INTELLIGENCE	FIRMWARE	CLOUD	DATABASE	DASHBOARD	CONNECTIVITY	OFFLINE
ESP32-S3 N16R8 2x I2S Mics OLED, Speaker, LEDs, Buttons	VAD + MFCC INT8 DS-CNN TFLite Micro ESP-DSP	ACWE + DSS RAAIS + ENF Adaptive Threshold Temporal Filtering	ESP-IDF FreeRTOS C / C++	FastAPI (Python) ASR + LLM + TTS REST API	Firebase Auth Cloud Firestore (NoSQL)	React / Vite HTML, CSS, JavaScript Three.js (3D)	Wi-Fi HTTPS / REST JSON	Flash Filesystem Smart Queue Auto Sync

1. FEASIBILITY AND VIABILITY

1. Technically Feasible

Built using an ESP32-S3, dual I2S microphones, TinyML wake-word detection (runs locally), OLED, speaker and Wi-Fi connectivity.

2. Low-Cost Hardware

Uses affordable and widely available components, making TRINETRA practical for prototype development and future deployment.

3. Edge AI Processing

Wake-word detection (TRINETRA) runs locally on the device using TinyML, reducing unnecessary cloud communication and improving response efficiency.

4. Hybrid Local + Ground Operation

Works in both online and offline conditions. Queries are processed locally during communication loss and synchronized with ground when link is restored.

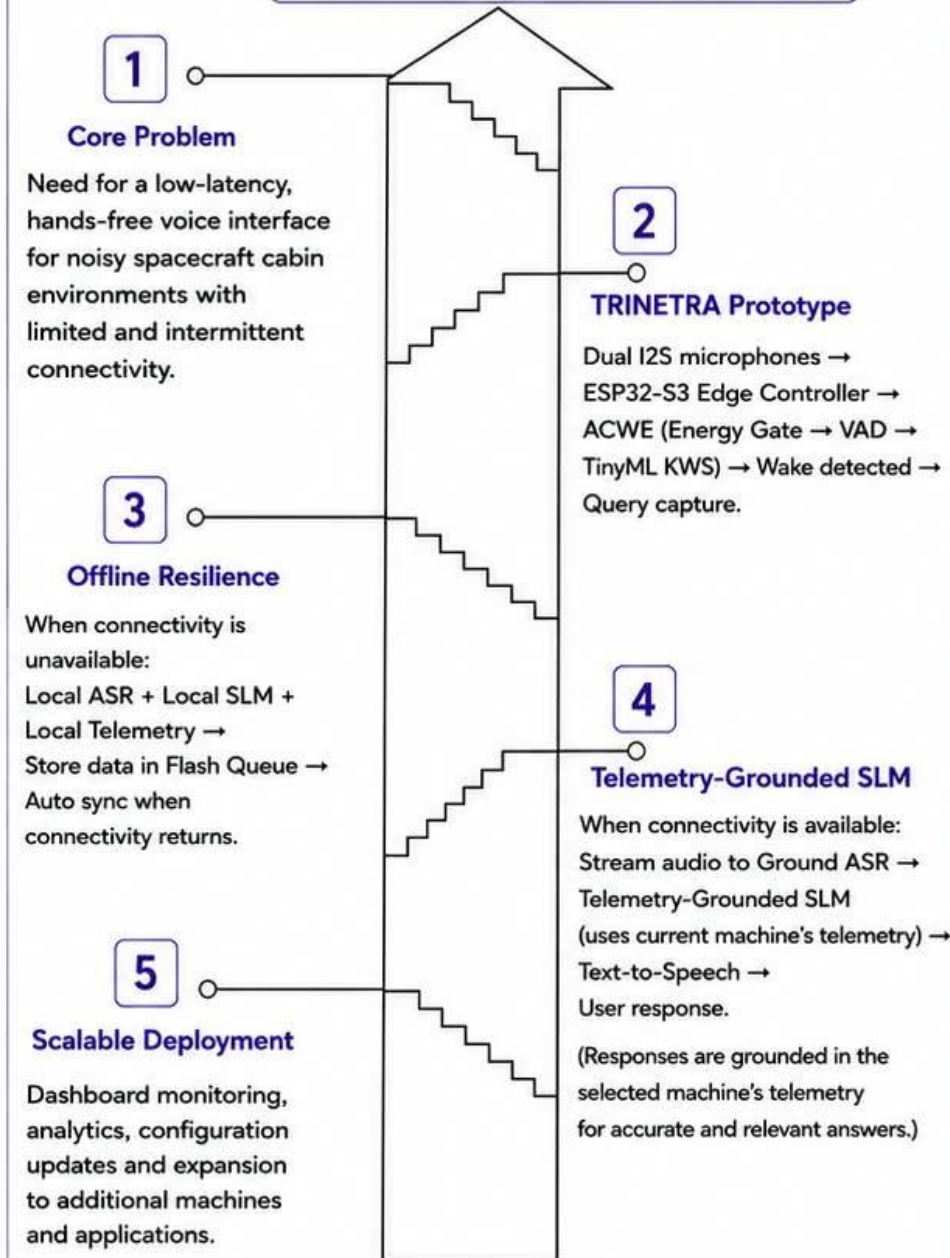
5. Energy & Resource Efficient

INT8 DS-CNN based cascaded pipeline with adaptive scheduling ensures idle CPU <10% and RAM <256 KB, activating heavy processing only when required.

6. Scalable Architecture

Modular Edge + Cloud architecture supports additional wake words, languages, dashboard features, telemetry tools and cloud services to be added later.

Business Aspects of TRINETRA



TELEMETRY DATA UTILIZATION

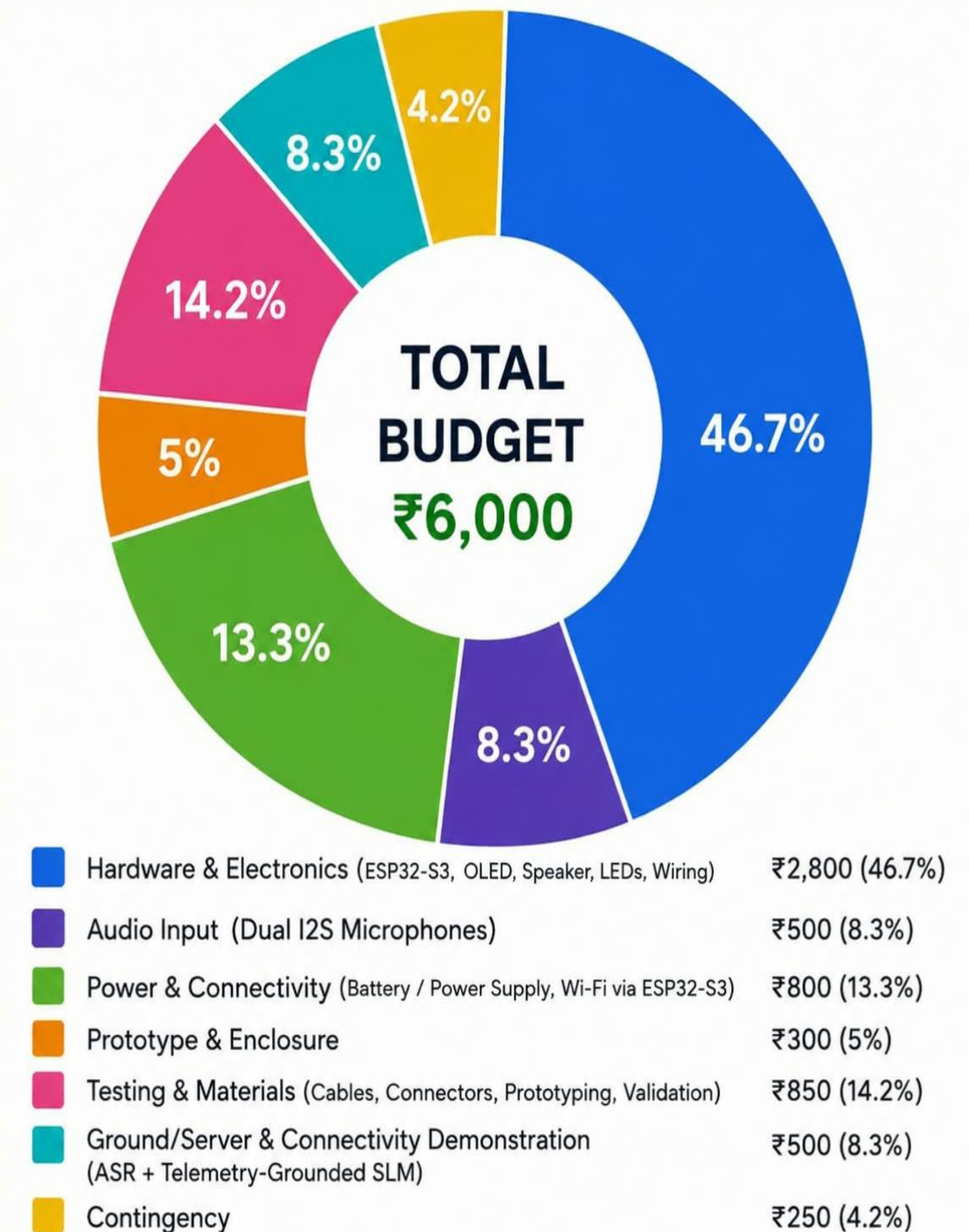
Live spacecraft telemetry (e.g., temperature, oxygen level, pressure, battery status, system health) is accessed through secure tools/APIs to provide real-time, context-aware responses.

Balancing Challenges with Innovative Solutions

Challenges	TRINETRA Solutions
 Noisy environments (telemetry: high noise level, SNR)	 Dual I2S microphones + Environmental Noise Fingerprinting (ENF)
 False wake-word triggers (false acceptance rate)	 Adaptive thresholds + cascaded wake-word detection + multi-frame validation
 Limited ESP32 resources (telemetry: CPU, RAM usage)	 Resource-Aware Adaptive Scheduling (RAAIS) + lightweight INT8 TinyML
 Poor internet connectivity (telemetry: connectivity status)	 Flash-based Smart Offline Queue + automatic sequential sync
 High latency for wake response (telemetry: end-to-end latency)	 Edge-first event-driven pipeline for wake detection ensuring low-latency activation
 Privacy concerns (telemetry: data handling logs)	 Local processing of ambient audio; only post-wake query audio sent when link is available
 High power consumption (telemetry: battery level, power usage)	 Low-power listening mode + event-driven activation + adaptive processing
 Need for real-time system data (telemetry: sensor values, system status)	 Telemetry APIs / Tool Calling for real-time spacecraft data retrieval



BUDGET DISTRIBUTION (Percentage)





BENEFITS OF THE SOLUTION



1. Mission & Crew Benefits

Enables hands-free voice interaction for astronauts, improving safety, efficiency, and situational awareness during critical mission operations.



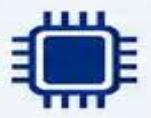
2. Cost & Resource Benefits

Low-cost ESP32-S3 and dual I2S microphones reduce hardware cost, power consumption, and bandwidth usage—ideal for tight mission budgets and limited resources.



3. Power & Efficiency Benefits

Low-power, energy-efficient design with adaptive scheduling and INT8 processing maximizes battery life and supports long-duration missions with minimal energy footprint.



4. Technical & Performance Benefits

RAAIS + ACWE + INT8 DS-CNN based cascaded wake detection ensures ultra-low latency (<1s), high accuracy, and robustness even in noisy, dynamic, and low-connectivity space environments.



5. Scalability & Future Benefits

Modular, adaptive architecture with telemetry-grounded intelligence and hybrid local + ground operation supports new wake words, multiple languages, and future mission services.

TARGET AUDIENCE
(Potential Impact)

1. Spacecraft & Mission Crew

- Hands-free control and information access during missions.
- Operates reliably in high-noise, microgravity, and confined environments.
- Reduces cognitive load and enhances mission safety.



2. Mission Control & Operations Teams

- Real-time voice queries for telemetry, system status, and alerts.
- Faster decision-making with accurate and contextual information.
- Improves coordination across distributed teams and centers.



3. Space Agencies, ISRO & Partners

- Affordable, indigenous, and scalable solution for space missions.
- Supports satellites, launch vehicles, habitats, and planetary missions.
- Strengthens self-reliance in voice AI for space applications.



4. Future Space Ecosystem

- Enables smart spacecraft, autonomous systems, and human-AI collaboration.
- Foundation for next-gen space missions, deep space exploration, and lunar/planetary bases.



IMPACTS



1. Social Impact

Improves accessibility and inclusivity in space missions, enabling seamless voice interaction for all crew members, including in high-stress and critical scenarios.



2. Technological Impact

Demonstrates an advanced, adaptive voice AI system combining RAAIS, ACWE, INT8 DS-CNN, telemetry-grounded SLM, and hybrid local + ground architecture for real-time, reliable performance in space.



3. Economic Impact

Reduces mission costs through low-cost hardware, optimized power usage, and minimal data transmission. Enables cost-effective large-scale deployment.



4. Environmental Impact

Energy-efficient and resource-aware processing minimizes power consumption and supports sustainable space exploration with reduced environmental footprint.



5. Long-Term Impact

Paves the way for intelligent, autonomous, and voice-enabled spacecraft, habitats, and exploration systems—accelerating innovation in space AI and human spaceflight.



KEY HIGHLIGHTS



Low Latency

Event-driven ACWE ensures wake response in < 1 second.



Energy Efficient

Cascaded detection, INT8 processing & adaptive scheduling minimize compute & battery use.



Privacy First

All ambient audio processed locally; only post-wake query audio sent to cloud when link is available.



Offline Resilient

Hybrid Local + Ground operation with Flash Queue; works without internet, syncs when online.



Edge AI Ready

Optimized for ESP32-S3 with TinyML (INT8 DS-CNN) for reliable on-device intelligence.



[ESP32-S3 I2S Documentation](#)

[TensorFlow Lite for Microcontrollers](#)

[Edge Impulse Documentation](#)

Comparision with Existing System

Feature	TRINETRA (ACWE)	Amazon Alexa (Gen 5)	ISRO Gaganyaan Communication System (AVPU)	Traditional Ground Communication (High Latency)
 Wake-Word Detection	✓	✓	N/A	N/A
 Processing Location	✓	✓	✓	✓
 Internet / Link Dependency (for Wake Detection)	✗	✗	N/A	N/A
 Internet / Link Dependency (Post-Wake Query Handling)	✓	✓	✓	✓
 Latency (Activation Response)	✓	✓	✗	✗
 Works in Poor Connectivity	✓	✗	✗	✗
 Energy Efficiency	✓	✗	✗	✗
 Target Environment	✓	✓	✓	✓
 Privacy (Local Processing)	✓	✗	✓	✗
 Primary Use Case	✓	✓	✓	✓
 Typical End-to-End Path	✓	✗	✗	✗
 Example Hardware / System	✓	✓	✓	✓